

Task-4 Classification Evaluation Report

1. Confusion Matrix Output

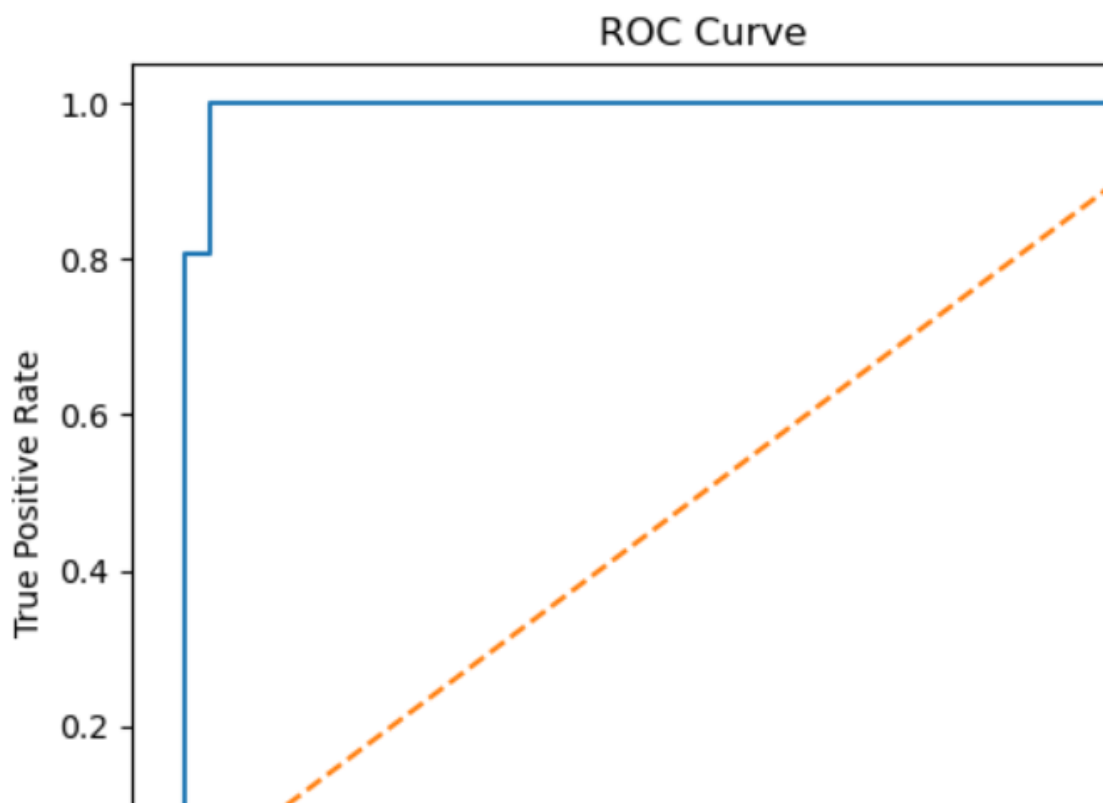
[[41, 1], [1, 71]]

True Negatives (TN)=41, False Positives (FP)=1, False Negatives (FN)=1, True Positives (TP)=71.

2. ROC Curve and AUC Score

AUC Score = 1.00

```
plt.title("ROC Curve")  
plt.legend()  
plt.show()
```



3. Model Comparison Results

Logistic Regression Accuracy: 0.98

Decision Tree Accuracy: 0.91

Logistic Regression outperformed the Decision Tree model.

4. Metric Selection

Accuracy, Precision, Recall, F1-Score and ROC-AUC were used to evaluate model performance.

5. Imbalanced Data Handling

`class_weight='balanced'` was used to demonstrate handling of imbalanced datasets.

6. Final Model Decision

Logistic Regression was selected as the final model due to its superior performance.